INCH-POUND
MIL-PRF-55342G
3 July 1997
SUPERSEDING
MIL-R-55342F
28 June 1995

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, FILM, CHIP, NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, SPACE LEVEL, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for nonestablished reliability (non-ER), established reliability (ER), and space level, fixed, film, chip resistors primarily intended for incorporation into surface mount applications. These resistors have a high degree of stability with respect to time under normal conditions. ER resistors covered by this specification have life failure rates (FR) ranging from 1 percent to 0.001 percent per 1,000 hours (see 1.2.7). These FRs are established at 60 percent confidence on the basis of life tests. Table I provides a summary of performance characteristics for these resistors.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). Resistors specified herein (see 3.1) are identified by a PIN which consists of the basic number and the associated specification. Each associated specification covers a different resistor style. The PIN provides information concerning the resistor characteristic, resistance value, resistance tolerance, termination material, and product designator. The PIN is in the following form:

* M55342	H	01	B	1E00	M
Specification indicating MIL-PRF-55342	Characteristic (see 1.2.3)	Specification sheet number indicating MIL-PRF-55342/1	Termination material (see 1.2.4)	Resistance and resistance tolerance (see 1.2.5 and 1.2.6)	Product level designator (see 1.2.7)

* A "D" prefix is used for metric performance specification.

1.2.2 Style. The style (see 3.1) is identified by the two-letter symbol "RM" followed by a four digit number; the letters identify fixed, film resistors, in a flat chip configuration.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications - Electronics Command and Fort Monmouth, ATTN: AMSEL-LC-LEO-E EP, Fort Monmouth, NJ 07703-5023, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
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1.2.3 Characteristic. The characteristic is identified by the single letter E, H, K, or M, in accordance with table I.

TABLE I. Characteristics.

Test or condition	Units	Symbol			
		E	H	K	M
Maximum ambient temperature at full wattage (see 3.6)	Temperature °C	70	70	70	70
Maximum ambient temperature at zero wattage (see 3.6)	Temperature °C	150	150	150	150
Thermal shock (see 3.9)	Maximum percent change in resistance (0.01 ohm additional allowed for measurement error)	±.1%	±.25%	±.5%	±.5%
Power conditioning (space level only) (see 3.10)		±0.2%	±0.25%	±0.5%	±0.5%
Low temperature operation (see 3.11)		±.1%	±.25%	±.25%	±.5%
Short time overload (see 3.12)		±.1%	±.1%	±.25%	±.5%
High temperature exposure (see 3.13)		±.1%	±.2%	±.5%	±1.0%
Resistance to bonding exposure (see 3.14)		±.2%	±.25%	±.25%	±.25%
Moisture resistance (see 3.15)	ppm/°C	±.2%	±.4%	±.5%	±.5%
Resistance temperature coefficient (see 3.16)		±25	±50	±100	±300
Life (see 3.17) 2,000 hours FR level		±.5% ±2.0%	±.5% ±2.0%	±.5% ±2.0%	±2.0% ±2.0%
Resistance tolerance ± percent (see table V)	± percent	0.1 1.0 2.0	0.1 1.0 2.0 5.0	0.1 1.0 2.0 5.0 10.0	1.0 2.0 5.0 10.0

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1.2.4 Termination material. The termination material designation will be in accordance with table II.

TABLE II. Termination materials.

Type	Material	Termination area	Code letters
Wire bondable	Gold	One surface Bonding pads	W 1/ P
Solderable 2/	Base metallization barrier metal, solder coated	Wraparound 3/	B
Epoxy bondable	Gold	Wraparound 3/	G
	Platinum gold	Wraparound 3/ One surface	U T 1/
	Palladium/silver or Platinum/silver	Wraparound 3/	C
	Palladium/silver or Platinum/silver	One surface	D 1/

1/ See 6.4.4.
2/ Solderable termination's will be pretinned for solder reflow operation and will meet the solderability test. The pretinning will be, as a minimum, on at least the bottom and ends of the chip and only those surfaces must meet the solderability test. (see figure 2)
3/ Wrap-around type will be illustrated on the associated specifications.

1.2.5 Resistance. The nominal resistance expressed in ohms is identified by four characters consisting of three digits and a letter. The letter is used simultaneously as a decimal point, multiplier, and a resistance tolerance designator. For resistance values:

a. Greater than or equal to 1 ohm but less than 1,000 ohms, the letters A, D, G, J, and M are used to represent a decimal point, depending on the resistance tolerance (see table III).

b. Greater than or equal to 1,000 ohms but less than 1 megohm, the letters B, E, H, K, and N are used to represent a decimal point, depending on the resistance tolerance (see table III).

c. Greater than or equal to 1 megohm, the letters C, F, T, L, and P are used to represent a decimal point, depending upon the resistance tolerance (see table III).

All digits preceding and following the letters (A, B, C, D, E, F, G, H, T, L, J, K, M, N, and P) of the group represent significant figures. The resistance value designators are shown in table III. Minimum and maximum resistance values will be as specified herein (see 3.1). The standard values for every decade will follow the sequence specified in table IV of MIL-PRF-55342 for resistance tolerances 1.0 percent, 2.0 percent, 5.0 percent, and 10.0 percent. The resistance values for tolerance 0.1 percent maybe any value within the limits specified (see 3.1), but it is preferred that values be chosen from the 0.1 percent column of table IV of MIL-PRF-55342.

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TABLE III. Designator of resistance values for resistance tolerances.

Designator for 0.1 percent tolerance	Resistance ohms
1A00 to 9A88 inclusive	1.00 to 9.88 inclusive
10A0 to 98A8 inclusive	10.0 to 98.8 inclusive
100A to 988A inclusive	100 to 988 inclusive
1B00 to 9B88 inclusive	1,000 to 9,880 inclusive
10B0 to 98B8 inclusive	10,000 to 98,800 inclusive
100B to 988B inclusive	100,000 to 988,000 inclusive
1C00 to 9C88 inclusive	1,000,000 to 9,880,000 inclusive
10C0	10,000,000
Designator for 1 percent tolerance	**Resistance ohms**
1D00 to 9D76 inclusive	1.00 to 9.76 inclusive
10D0 to 97D6 inclusive	10.0 to 97.6 inclusive
100D to 976D inclusive	100 to 976 inclusive
1E00 to 9E76 inclusive	1,000 to 9,760 inclusive
10E0 to 97E6 inclusive	10,000 to 97,600 inclusive
100E to 976E inclusive	100,000 to 976,000 inclusive
1F00 to 9F76 inclusive	1,000,000 to 9,760,000 inclusive
10F	10,000,000
Designator for 2 percent tolerance	**Resistance ohms**
1G00 to 9G10 inclusive	1.00 to 9.10 inclusive
10G0 to 91G0 inclusive	10.0 to 91.0 inclusive
100G to 910G inclusive	100 to 910 inclusive
1H00 to 9H10 inclusive	1,000 to 9,100 inclusive
10H0 to 91H0 inclusive	10,000 to 91,000 inclusive
100H to 910H inclusive	100,000 to 910,000 inclusive
1T00 to 9T10 inclusive	1,000,000 to 9,100,000 inclusive
10T0	10,000,000
Designator for 5 percent tolerance	**Resistance ohms**
1J00 to 9J10 inclusive	1.00 to 9 .10 inclusive
10J0 to 91J0 inclusive	10.0 to 91.0 inclusive
100J to 910J inclusive	100 to 910 inclusive
1K00 to 9K10 inclusive	1,000 to 9,100 inclusive
10K0 to 91K0 inclusive	10,000 to 91,000 inclusive
100K to 910K inclusive	100,000 to 910,000 inclusive
1L00 to 9L10 inclusive	1,000,000 to 9,100,000 inclusive
10L0	10,000,000
Designator for 10 percent tolerance	**Resistance ohms**
1M00 to 9M10 inclusive	1.00 to 8.20 inclusive
10M0 to 91M0 inclusive	10.0 to 82.0 inclusive
100M to 910M inclusive	100 to 820 inclusive
1N00 to 9N10 inclusive	1,000 to 8,200 inclusive
10N0 to 91N0 inclusive	10,000 to 82,000 inclusive
100N to 910N inclusive	100,000 to 820,000 inclusive
1P00 to 9P10 inclusive	1,000,000 to 8,200,000 inclusive
10P0	10,000,000

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TABLE IV. Standard resistance values for the 10 to 100 decade.

0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0
10.00	10.00	10.00	10.00	20.00	20.00	20.00		37.90			
10.10				20.30				38.30	38.30		
10.20	10.20			20.50	20.50			38.80			
10.40				20.80						39.00	39.00
10.50	10.50			21.00	21.00			39.20	39.20		
10.60				21.30				39.70			
10.70	10.70			21.50	21.50			40.20	40.20		
10.90				21.80				40.70			
11.00	11.00	11.00				22.00	22.00	41.20	41.20		
11.10				22.10	22.10			41.70			
11.30	11.30			22.30				42.20	42.20		
11.40				22.60	22.60			42.70			
11.50	11.50			22.90						43.00	
11.70				23.20	23.20			43.20	43.20		
11.80	11.80			23.40				43.70			
12.00		12.00	12.00	23.70	23.70			44.20	44.20		
12.10	12.10			24.00		24.00		44.80			
12.30				24.30	24.30			45.30	45.30		
12.40	12.40			24.60				45.90			
12.60				24.90	24.90			46.40	46.40		
12.70	12.70			25.20				47.00		47.00	47.00
12.90				25.50	25.50			47.50	47.50		
13.00	13.00	13.00		25.80				48.10			
13.20				26.10	26.10			48.70	48.70		
13.30	13.30			26.40				49.30			
13.50				26.70	26.70			49.90	49.90		
13.70	13.70					27.00	27.00	50.50			
13.80				27.10						51.00	51.00 1/
14.00	14.00			27.40	27.40			51.10	51.10		
14.20				27.70				51.70			
14.30	14.30			28.00	28.00			52.30	52.30		
14.50				28.40				53.00			
14.70	14.70			28.70	28.70			53.60	53.60		
14.90				29.10				54.20			
15.00	15.00	15.00	15.00	29.40	29.40			54.90	54.90		
15.20				29.80				55.60			
15.40	15.40					30.00				56.00	56.00
15.60				30.10	30.10			56.20	56.20		
15.80	15.80			30.50				56.90			
16.00		16.00		30.90	30.90			57.60	57.60		
16.20	16.20			31.20				58.30			
16.40				31.60	31.60			59.00	59.00		
16.50	16.50			32.00				59.70			
16.70				32.40	32.40			60.40	60.40		
16.90	16.90			32.80				61.20			
17.20						33.00	33.00	61.90	61.90		
17.40	17.40			33.20	33.20					62.00	
17.60				33.60				62.60			
17.80	17.80			34.00	34.00			63.40	63.40		
18.00		18.00	18.00	34.40				64.20			
18.20	18.20			34.80	34.80			64.90	64.90		
18.40				35.20				65.70			
18.70	18.70			35.70	35.70			66.50	66.50		
18.90						36.00		67.30			
19.10	19.10			36.10						68.00	68.00
19.30				36.50	36.50			68.10	68.10		
19.60	19.60			37.00				69.00			
19.80				37.40	37.40						

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TABLE IV. Standard resistance values for the 10 to 100 decade - Continued.

0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0
69.80	69.80			79.60				88.70	88.70		
70.60				80.60	80.60			89.80			
71.50	71.50			81.60				90.90	90.90		
72.30						82.00	82.00			91.00	
73.20	73.20			82.50	82.50			92.00			
74.10				83.50				93.10	93.10		
75.00	75.00	75.00	1/ 75.00	84.50	84.50			94.20			
75.90				85.60				95.30	95.30		
76.80	76.80			86.60	86.60			96.50			
77.70				87.60				97.60	97.60		
78.70	78.70							98.80			

1/ Inactive for new design

1.2.6 Resistance tolerance and multipliers. The resistance tolerance and multipliers are identified by a single letter in accordance with table V.

TABLE V. Resistance tolerance and multipliers.

Resistance tolerance (±)	Multiplier	Symbol
0.1	X1	A
0.1	X1,000	B
0.1	X1,000,000	C
1.0	X1	D
1.0	X1,000	E
1.0	X1,000,000	F
2.0	X1	G
2.0	X1,000	H
2.0	X1,000,000	T
5.0	X1	J
5.0	X1,000	K
5.0	X1,000,000	L
10.0	X1	M
10.0	X1,000	N
10.0	X1,000,000	P

1.2.7 Product level designator. The product level designator is specified in table VI by a single letter which identifies non-ER, ER or space level resistors.

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TABLE VI. Product level designator.

Product level designator	Product level
C	Non-ER
M	1/ 1.0
P	1/ 0.1
R	1/ 0.01
S	1/ 0.001
T	Space level

1/ FR level in percent per 1000 hours

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

MMM-A-1931 - Adhesive, Epoxy, Silver Filled, Conductive.

DEPARTMENT OF DEFENSE

MIL-I-46058 - Insulating, Compound, Electrical (For coating printed circuit assembles).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 - Electronic and Electrical Component Parts, Test Methods for.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Standard Practice for Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications.
MIL-STD-883 - Microelectronics, Test Methods and Procedures for.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(See supplement 1 for list of associated specifications.)

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-554-1 - Assessment of Outgoing Nonconforming Levels in Parts Per Million (ppm). (DoD adopted).
EIA-557 - Statistical Process Control Systems (DoD adopted).

(Applications for copies should be addressed to Electronic Industries Association (EIA), 2500 Wilson Boulevard, Arlington, VA 22201-3834.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B545 - Standard Specification for Electrodeposited Coating of Tin.
ASTM E595 - Materials from Outgassing in a Vacuum Environment, Total Mass Loss and Collected Volatile Condensable, Standard Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NASA 1124 - Outgassing Data for Selecting Spacecraft Materials.

(Copies of NASA 1124 Publication are available from NASA / Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specifications. The individual item requirements shall be as specified herein and in accordance with the applicable associated specification. In the event of any conflict between the requirements of this specification and the associated specifications, the latter shall govern (see 6.2).

3.2 Qualification. Resistors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award. In addition, the manufacturer shall obtain certification from the qualifying activity that the product assurance requirements of 4.2.1 have been met and are being maintained. Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturer are listed in the QPL.

3.3 QPL system. The manufacturer shall establish and maintain a QPL system for parts covered by this specification. Requirements for this system are specified in MIL-STD-790 (all product levels) and MIL-STD-690 (ER parts only). In addition, the manufacturer shall also establish a Statistical Process Control (SPC) and Part Per Million (ppm) system that meets the requirements as described in 3.3.1 and 3.3.2 respectively.

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3.3 1 SPC system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish an SPC system that meets the requirements of EIA-557. Typical manufacturing processes for application of SPC include formation of resistance elements, trimming, and termination. In addition, the manufacturer shall demonstrate resistance temperature characteristic (RTC) control in the process.

3.3.2 PPM system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish an ppm system for assessing the average outgoing quality of lots in accordance with EIA-554-1 and 4.6.4. Data exclusion, in accordance with EIA 554-1, may be used with approval of the qualifying activity. The ppm system shall identify the ppm rate at the end of each month and shall be based on a six month moving average. PPM-2 and dc resistance, shall be assessed for each style. Style reporting may include both non-ER and ER style combinations. Due to low production volume, ppm assessment does not apply to space level lots.

3.4 Materials. Materials shall be used which will enable the resistors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Interface and physical dimension requirements. Resistors shall meet the interface and physical dimensions specified (see 3.1).

3.5.1 Termination. Unless otherwise specified, both termination's (areas) shall be available on one surface of the body of the resistor chip (see 1.2.4).

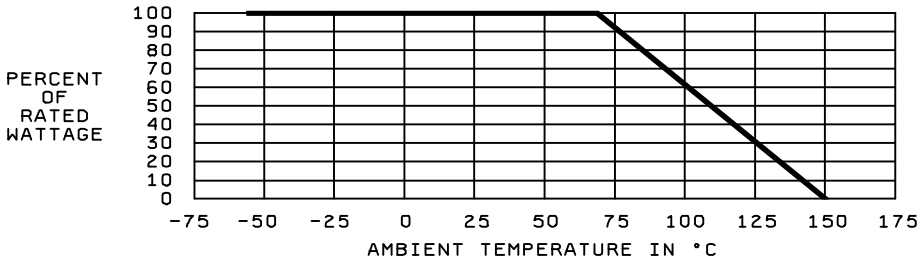
3.5.2 Barrier metallization. The barrier metallization for the B termination (base metallization barrier metal, solder coated) shall be nickel. The metallization shall be a minimum of 50 microinches.

3.5.2.1 Terminal substitution data. For existing design and logistical support of the deleted termination R (solderable pretinned wraparound termination's), termination B (solderable base metallization barrier metal, solder coated) shall be used as a substitute for termination R.

3.5.3 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.4.5). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

3.6 Power rating. The resistors shall have a power rating based on continuous full-load operation at an ambient temperature of 70ºC. For operation at temperatures in excess of 70ºC the load shall be derated in accordance with figure 1.

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NOTE: This curve indicates the percentage of nominal wattage to be applied at temperature higher than 70ºC. This curve applies only to units mounted on a board (see 4.8.1.3); however, at no time shall the applied voltage exceed the maximum for each style (see 3.1).

FIGURE 1. Derating curve for high ambient temperatures.

3.7 Voltage rating. Resistors shall have a rated dc continuous working voltage or an approximate sinewave root-mean-square (rms) continuous working voltage corresponding to the wattage (power) rating, as determined from the following formula:

$$E = \sqrt{PR}$$

Where:

E = Continuous rated dc or rms working voltage in volts
P = Rated wattage in watts
R = Nominal resistance in ohms

In no case shall the rated voltage be greater than the applicable maximum voltage (see 3.1).

3.8 DC resistance. When resistors are tested as specified in 4.8.2, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.5).

3.9 Thermal shock.

3.9.1 Thermal shock (failure rate levels (FRL) M, P, R, and S). When resistors are tested as specified in 4.8.3, there shall be no evidence of mechanical damage; the change in resistance shall be within the specified initial resistance tolerance of the nominal resistance for group A tests.

3.9.2 Thermal shock (space level only). When resistors are tested as specified in 4.8.3, there shall be no evidence of mechanical damage; the combined change in resistance for thermal shock and power conditioning during group A inspection shall not exceed the values listed in table I for power conditioning. For the 100 cycle qualification test, the change in resistance shall not exceed the values listed in table I for the power conditioning.

3.10 Power conditioning (space level only). When resistors are tested as specified in 4.8.4, there shall be no evidence of mechanical damage; the combined change in resistance, for thermal shock and power conditioning, shall not exceed the values listed in table I for power conditioning.

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3.11 Low temperature operation. When resistors are tested as specified in 4.8.5, there shall be no evidence of mechanical damage; the change in resistance between the initial and the final measurement at 25ºC ±5ºC, shall not exceed the value specified in table I.

3.12 Short time overload. When resistors are tested as specified in 4.8.6, there shall be no evidence of arcing, burning, or charring, the change in resistance shall not exceed the value specified in table I.

3.13 High temperature exposure. When resistors are tested as specified in 4.8.7, there shall be no evidence of mechanical damage and the change in resistance shall not exceed the value specified in table I.

3.14 Resistance to bonding exposure. When resistors are tested as specified in 4.8.8, there shall be no evidence of mechanical damage, no demetallization or leaching, and the change in resistance shall not exceed the value specified in table I.

3.15 Moisture resistance. When resistors are tested as specified in 4.8.9, there shall be no evidence of mechanical damage; the change in resistance between the initial and final measurements shall not exceed the value specified in table I.

3.16 Resistance temperature characteristic. When resistors are tested as specified in 4.8.10, the resistance temperature characteristic, at each of the temperatures specified in table XIV referred to room ambient temperature, shall not exceed the values specified in table I.

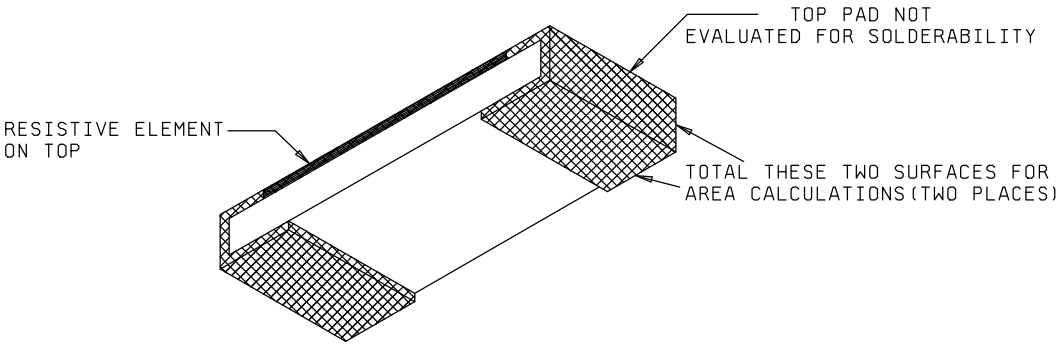
3.17 Life.

3.17.1 Qualification. When resistors are tested as specified in 4.8.11, there shall be no evidence of mechanical damage. The change in resistance between the initial measurement and any succeeding measurements up to and including 2,000 hours, shall not exceed the value specified in table I.

3.17.2 FRL determination (ER only). When resistors are tested as specified in 4.8.11, there shall be no evidence of mechanical damage to the resistance element, or enclosure. The change in resistance between the initial measurement and any of the succeeding measurements shall not exceed the value specified in table I. This single failure criteria shall be applicable to all measurements during the life test for purposes of determining FRL qualification and is applicable as a parallel requirement with 3.17.1 to measurements made during the life test specified for qualification inspection.

3.18 Solderability (applicable to termination's B). When resistors are tested as specified in 4.8.12, the immersed metallized surface shall be at least 95 percent covered with a new clean smooth coating and shall exhibit no demetallization or leaching of the terminal areas. The remaining 5 percent may contain only small pinholes or rough spots; these shall not be concentrated in one area. In case of dispute, the percentage of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area. The top surface of the solder pad need not be considered in the evaluation for solderability (see figure 2). For coverage calculations, defective areas on each termination shall be compared to the total area of the end and bottom surfaces on the same termination.

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NOTES:
1. Top solder pad need not be evaluated for solderability.
2. Resistive element on top.
3. Ends and bottom are surfaces to be evaluated.

FIGURE 2. Solderability coverage.

3.19 Mounting integrity.

3.19.1 Solder mounting integrity (applicable to termination B). When resistors are tested as specified in 4.8.13.1, there shall be no evidence of mechanical damage.

3.19.2 Bondable mounting integrity (applicable to termination's C, D, G, T, and U). When resistors are tested as specified in 4.8.13.2, there shall be no evidence of mechanical damage.

3.19.3 Wire bonding integrity (applicable to termination's W and P). When resistors are tested as specified in 4.8.13.3, there shall be no evidence of mechanical damage.

3.20 Resistance to solvents (applicable to resistors that are marked with color dots or screen printed). When resistors are tested as specified in 4.8.14, there shall be no evidence of mechanical damage and the marking shall remain legible. The color dot marking shall be in accordance with MIL-STD-1285.

3.21 Marking legibility.

3.21.1 Marking legibility test (applicable to laser marked resistors). When resistors are tested as specified in 4.8.15, the marking shall remain legible.

3.22 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.23 Visual inspection. Resistors shall be inspected as specified in 4.8.1, to verify that the interface, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.5.1 to 3.5.3 inclusive, and appendix B).

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3.24 Outgassing (space level only). When examined as specified in 4.8.16, the samples shall meet the following requirements:

a. Total Mass Loss (TML) shall not exceed 1 percent.

b. Volatile Condensable Material (VCM) shall not exceed 0.1 percent

3.24.1 Outgassing test data. Data listed in NASA Publication 1124 may be used in lieu of actual test data for applicable materials.

3.25 Marking. Marking of individual chip resistors is not required but is permitted. A noncorrosive label containing the following information shall be applied to each unit package. PIN, "JAN" marking, date code, source code, and manufacturer's production lot code (see 4.6.2.2). Date and source code shall be in accordance with MIL-STD-1285. The following is an example of the complete marking:

12345 - CAGE
7343J - Date code and JAN marking
M55342H01 - PIN
S10A5M - PIN
- Manufacturer's production lot code

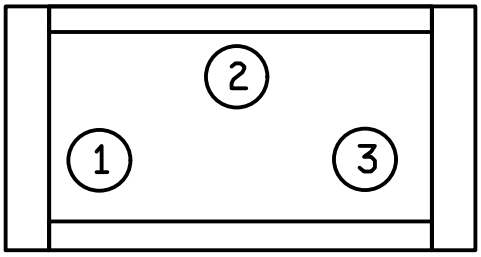
The date code shall be the date of the final assembly operation for the production lot as defined in 4.6.2.2. The common manufacturing record shall include the same date code as that placed on the parts covered by the record. At the option of the manufacturer, resistors may be marked (i.e., laser, color dot, screen printed, etc.). The marking process must be approved by the qualifying activity based on testing specified herein. The marking shall remain legible after all tests.

3.25.1 Marking (color dot, screen print, etc.).

3.25.1.1 Three dot option (designates 2 percent, 5 percent, and 10 percent values). An example of three dot marking option shall be as in the following example:

1st dot - 1st significant digit
2nd dot - 2nd significant digit
3rd dot - multiplier (reference MIL-STD-1285)

example: 330k ohm orange
orange yellow



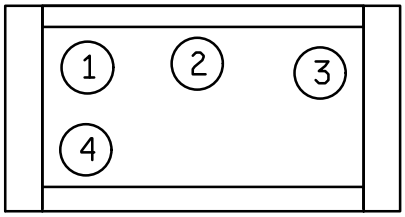
NOTE: Abbreviated marking due to limited space takes precedence over PIN format.

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3.25.1.2 Four dot option (designates 1 percent values). An example of four dot marking option shall be as in the following example:

1st dot	-	1st significant digit
2nd dot	-	2nd significant digit
3rd dot	-	3rd significant digit
4th dot	-	multiplier (reference MIL-STD-1285)

example: 30.1k ohm orange black brown
red



NOTE: Abbreviated marking due to limited space takes precedence over PIN format.

3.25.2 Laser marking/screen printing/stamping. An example of the code used for laser marking, screen printing, or stamping shall be as shown in the following example.

a. Three digit code:

330k ohms shall be coded 334

12 ohms shall be coded 120

b. Four digit code (includes tolerance and shall follow the table III format):

30.1k ohms one percent shall be coded 30E1

18.2 ohms one percent shall be coded 18D2

NOTE: Abbreviated marking due to limited space takes precedence over PIN format.

3.25.3 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable associated specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable associated specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated specifications, the manufacturer shall remove completely the military part number and the "JAN" or "J" from the sample tested and also from all items represented by the sample. The “JAN” or “J” certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 1,586,261 for the certification mark “J”.

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3.25.4 Supplying to higher FRLs and product levels. A manufacturer may supply to all higher product levels to which he is qualified. Parts qualified to lower product levels (i.e., more stringent), with procuring agency approval, are substitutable for higher product levels as long as the label reflects the original product level ordered (see table VII). Parts that are physically marked shall not be remarked unless specified in the contract or order (see 6.2).

3.25.5 Supplying to lower resistance temperature characteristics and or lower resistance tolerances. Parts qualified to lower resistance temperature characteristics and/or lower resistance tolerance levels are substitutable, with procuring activity approval, for higher resistance temperature characteristic or higher tolerance levels as long as the label reflects the original resistance temperature characteristic and/or resistance tolerance (see table VII). Parts that are physically marked shall not be remarked unless specified in the contract or order (see 6.2).

3.26 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, and 3.25 as applicable, and be free from other defects that will affect life, serviceability, or appearance, and shall pass the visual inspection as specified on figure 6 through figure 10 in Appendix B.

TABLE VII. Substitution data.

Product level	Acceptable FR substitute
T (space) S (0.001) R (0.01) P (0.1) M (1.0) C (non-ER)	----- T T, S T, S, R, T, S, R, P T, S, R, P, M
Resistance tolerance	Acceptable resistance tolerance substitute
0.1 1.0 2.0 5.0 10.0	---- 0.1 0.1, 1.0 0.1, 1.0, 2.0 0.1, 1.0, 2.0, 5.0
Characteristic	Acceptable characteristic substitute
E H K M	--- E --- K

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Conformance inspection (see 4.6).

d. Periodic inspection (see 4.7)

4.2 Reliability and quality.

4.2.1 QPL system. The manufacturer shall establish and maintain a QPL system as described in 3.3. Evidence of such compliance is a prerequisite for qualification and retention of qualification.

4.2.2 SPC program. A SPC program shall be maintained in accordance with EIA-557. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.3 Inspections conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the general requirements of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture resistance temperature cycling tests.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3).

4.4.1 Sample. The number of sample units comprising a sample of resistors to be submitted for qualification inspection shall be as specified in appendix A to this specification. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production and which have been subjected to and passed the requirements of group A inspection (see 4.6.3). Qualification shall not be granted if group A inspection requirements are not complied with.

4.4.2 Test routine. Sample units shall be subjected to the qualification inspection specified in table VIII in the order shown. All sample units, with the exception of ten sample units for group V inspection, shall be subjected to the inspection of group I. The 182 sample units shall be divided as specified in table VIII for group II, group III, group IV, and group VI and subjected to the inspection for their particular group. Ten sample units shall be submitted to group VII, if applicable. Sample sizes and extent of qualification for characteristics shall be specified in the appendix to this specification.

4.4.3 Defective. Defectives in excess of those allowed in table VIII shall be cause for refusal to grant qualification.

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4.4.4 FRL verification (ER only).

4.4.4.1 Procedure. FR qualification shall be in accordance with the general requirements of MIL-STD-690 and the following details:

a. Procedure I: Qualification at the initial FR level. Level M (1.0 percent) of FRSP-60 shall apply. Sample units shall be subjected to the qualification inspection specified in group IV, table VIII (see 4.4.2). Entire life test sample shall be continued on test to 10,000 hours as specified in 3.17.1 upon completion of the 2,000 hour qualification.

b. Procedure II: Extension of qualification to lower FR levels. To extend qualification to the R (0.01 percent) and S (0.001 percent) FR levels, two or more styles of similar construction may be combined.

c. Procedure III: Maintenance of FR level qualification. Maintenance period A of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.7).

4.5 Verification of qualification. Every 6 months the manufacturer shall provide verification of qualification to the qualifying activity. Continuation of qualification is based on meeting the following requirements.

a. MIL -STD-790 program.

b. Design of resistor has not been modified.

c. Lot rejection for group A (subgroup 2 and subgroup 4) does not exceed 5 percent or one lot, whichever is greater.

d. Lot rejection for group B inspection does not exceed 5 percent or one lot, whichever is greater.

e. Periodic group C inspection requirements are met.

f. FRLs are maintained as required.

g. PPM assessment. (NOTE: Grouping of styles is permitted.

h. Continued qualification to the non-ER level (C) shall be based on continued maintenance of qualification for the ER part (minimum P FRL maintenance).

i. Continued qualification to the space level (T) shall be based on maintaining minimum ER FRL of R or S and maintenance of the test capability and lot control system for testing space level resistors.

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TABLE VIII. Qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Certification requirements				
Outgassing (space level only)	3.24	4.8.16	-----	----
Group I				
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.23, and 3.25	4.8.1	All sample units except group V	0
DC resistance	3.8	4.8.2		
Group II				
Resistance to bonding exposure	3.14	4.8.8	10 highest	1
Resistance temperature characteristic	3.16	4.8.10	10 critical	
Low temperature operation	3.11	4.8.5	10 lowest	
Short time overload	3.12	4.8.6	30	
High temperature exposure	3.13	4.8.7		
Group III				
Resistance to bonding exposure	3.14	4.8.8	10 highest	1
Moisture resistance	3.15	4.8.9	10 critical	
			10 lowest	
			(30)	
Group IV				
Life	3.17	4.8.11.1	34 highest	1
			34 critical	
			34 lowest	
			(102)	
Group V				
Solderability 1/	3.18	4.8.12	10 any value	1
Group VI				
Solderable mounting integrity 1/	3.19.1	4.8.13.1	10 any value	1
Bondable mounting integrity 2/	3.19.2	4.8.13.2	10 any value	
Wire bonding integrity 3/	3.19.3	4.8.13.3	10 any value	
			(30)	
Group VII				
Resistance to solvents 4/	3.20	4.8.14	10 any value	0
Marking legibility test (laser marking) 5/	3.21.1	4.8.15	10 any value	
			(20)	
Group VIII (space level only)				
Thermal shock (100 cycles)	3.9	4.8.3	10 highest	0
			10 critical	
			10 lowest	
			(30)	

1/ Applicable to termination B.
2/ Applicable to termination C, D, G, T and U.
3/ Applicable to termination W and P.
4/ Test applicable for screen printed or color dot parts.
5/ Test applicable for laser marked parts.

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4.6 Conformance inspection.

4.6.1 Inspection of product for delivery.

4.6.1.1 Non-ER resistors. The manufacturer's inspection system shall be used for preparation for delivery.

4.6.1.2 ER resistors. Inspection of product for delivery shall consist of group A and group B inspections. Group B inspection for preparation of delivery is not required when the qualifying activity has allowed group B testing to be performed annually (see table XI).

4.6.1.3 Space level. Inspection of product for delivery shall consist of the group A and B inspection.

4.6.2 Inspection and production lot.

4.6.2.1 Inspection lot. An inspection lot shall consist of resistors of the same style, characteristic, and termination type and manufactured under essentially the same process and conditions with the same basic materials and offered for inspection at one time. Non-ER, ER, and space level lots shall be kept separate.

4.6.2.2 Production lot (ER and space level only). A production lot shall consist of all resistors of the same style, nominal resistance value, resistance tolerance, resistance temperature characteristic, voltage rating, and terminal type. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle. In addition, for space level production lots only; the deposition/metallization shall be restricted to substrates produced for a single resistor style; the substrates shall be processed in a single continuous metallization run on the same equipment and the raw material used in the metallization process shall be from the same raw material lot. Non-ER, ER, and space level lots shall be kept separate.

4.6.2.3 Optional production lot control (characteristic E and characteristic K only). A manufacturer that has a single design that meets all the requirements of both characteristic E and characteristic K may, with qualifying activity approval, use a different control system. All products shall be processed as E characteristic but may be subsequently supplied as K characteristic provided the manufacturer marks the unit packages accordingly and performs all other tests for characteristic K.

4.6.3 Group A inspection.

4.6.3.1 Non-ER resistors. The manufacturer shall establish and maintain an inspection system to verify that resistors meet dc resistance, visual/mechanical, and solderability requirements. In-line or process controls may be part of such a system. The inspection system shall also include criteria for lot rejection and corrective actions. The inspection system shall be verified under the overall MIL-STD-790 QPL system. NOTE: Since the non-ER (C level) is the ER design without the mandatory conformance inspection and FRL assessment, this product is still expected meet the environmental qualification type requirements (e.g., resistance to bonding exposure, moisture resistance, high temperature exposure, etc.).

4.6.3.2 ER and space level resistors. Group A inspection shall consist of the examinations and tests specified in table IX.

4.6.3.2.1 Subgroup 1 test (precap visual). The subgroup 1 test shall be performed on a production lot basis. Thirteen samples shall be subjected to subgroup 1. In the event of one or more failures, the lot shall be rejected.

4.6.3.2.1.1 Rejected lots. In the event of one or more defects, and the production lot is rejected, the rejected production lot(s) shall be submitted to a 100 percent precap visual inspection as specified in 3.23. Resistors that pass precap visual inspection are available to continue group A testing.

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4.6.3.2.2 Subgroup 2 tests. Subgroup 2 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup, shall be removed from the lot. Lots having more than 5 percent total rejects due to exceeding the specified resistance tolerance change limit shall not be furnished on contracts.

4.6.3.2.2.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in subgroup 2, table IX, as the final step of his production process, group A, subgroup 2 inspection may be waived. Authority to waive the subgroup 2 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup 2. Test conditions shall be equal to or more stringent than those specified for subgroup 2 tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same as, or more stringent than, those specified herein.

d. The lot rejection criteria are the same as, or more stringent than, those specified herein.

e. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification to and concurrence from the qualifying activity.

TABLE IX. Group A inspection (ER and space level).

Inspection	Requirement paragraph	Method paragraph	Sampling procedure
Subgroup 1			
Precap visual inspection 1/	3.23	4.8.1	4.6.3.2.1
Subgroup 2			
Thermal shock 2/	3.9	4.8.3	4.6.3.2.2
Power conditioning (space level only)	3.10	4.8.4	
DC resistance	3.8	4.8.2	
Subgroup 3			
Visual inspection	3.1, 3.4, 3.5 through 3.5.3 inclusive, and 3.23	4.8.1	4.6.3.2.4
Subgroup 4			
Solderability 3/	3.18	4.8.12	4.6.3.2.5
Subgroup 5 4/			
Resistance to solvents 5/	3.20	4.8.14	4.6.3.2.6
Marking legibility 6/	3.21	4.8.15	

See footnotes at end of table.

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TABLE IX. Group A inspection (ER and space level) - Continued.

1/ Precap visual inspection shall be performed when the resistance element can not be inspected after passivation/glassivation in subgroup 3.
2/ If the manufacturer can demonstrate that the FRL of the parts can be maintained without performing the thermal shock screen, this test, with approval of the preparing activity and qualifying activity, may be deleted. If the design, material, construction, or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the test. As a minimum, for initial consideration of this option, the manufacturer shall demonstrate and validate using data equivalent to M FR qualification, that the test does not affect the FR of the resistor. Upon deletion of the test, the manufacturer must continue to maintain their existing FRL based on testing of parts without the thermal shock screening.
3/ The manufacturer may request the deletion of the Subgroup 4 solderability test, provided an in-line or process control system for assessing and assuring the solderability of terminations can be validated and approved by the qualifying activity. Deletion of the test does not relieve the manufacturer from meeting this test requirement in case of dispute. If the design, material, construction, or processing of the part is changed or if there are any quality problems, the qualifying activity may require resumption of the test.
4/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity, can be performed on a annual basis. If the design, material, construction, or processing of the part is changed or , if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.
5/ Test applicable for color dot or screen printed parts.
6/ Test applicable for laser marked parts.

4.6.3.2.4 Subgroup 3 (visual inspection). The subgroup 3 test shall be performed on an inspection lot basis for ER parts and a production lot basis for space level parts. A sample of parts shall be randomly selected in accordance with table X. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defectives, a new sample of parts shall be randomly selected in accordance with table X. If one or more defects are found in this second sample for the sample quality characteristic, the lot shall be rejected and shall not be supplied to this specification.

4.6.3.2.5 Subgroup 4 (solderability) test. The subgroup 4 test shall be performed on an inspection lot basis for ER parts and on a production lot basis for space level parts. A sample shall be randomly selected from each lot in accordance with table X. As an option, the manufacturer may use electrical rejects from the subgroup 2 tests for all or part of the sample. If there are one or more defects, the lot shall be considered to have failed.

4.6.3.2.5.1 Rejected lots. In the event of one or more defects, the inspection lot is rejected. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 3.18. An additional sample shall be randomly selected from each production lot in accordance with table X. If the lot fails this solderability test, the lot may be reworked and retested. Any individual production lot that fails the solderability retest shall be considered to have failed and that lot will not be supplied to the requirements of the specification.

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TABLE X. Sampling plans for subgroup 3 and subgroup 4 and ppm assessment.

Lot size	Sample size for subgroup 3	Sample size for subgroup 4	Sample size for ppm assessment
1 to 5	100%	100%	100%
6 to 13	100%	5	100%
14 to 125	13	5	100%
126 to 150	13	5	125
151 to 280	20	5	125
281 to 500	29	5	125
501 to 1,200	34	5	125
1,201 to 3,200	42	5	125
3,201 to 10,000	50	8	294
10,001 to 35,000	60	13	294
35,001 to 150,000	74	20	345
150,001 to 500,000	90	20	435
500,001 and over	102	20	

4.6.3.2.5.2 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

4.6.3.2.6 Subgroup 5. Eight samples shall be selected randomly from each inspection lot and subjected to the subgroup 5 tests. The manufacturer may use electrical rejects from the subgroup 2 screening tests for all or part of the samples to be used for the subgroup 5 testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.3.2.6.1 Rejected lots. In the event of one or more defects, the inspection lot is rejected. Each production lot that was used to form the failed inspection lot shall be individually submitted to the subgroup 5 test as required in 3.20 or 3.21. An additional eight samples shall be randomly selected from each production lot and shall be tested. If there are one or more defects in the production lot sample, that production lot shall be considered to have failed and that product will not be supplied to the requirements of this specification. Individual production lots that pass the minimum marking test are available for shipment.

4.6.4 PPM assessment (non-ER and ER). The manufacturer shall establish and maintain a system for assessing the average outgoing quality in ppm of lots supplied to this specification. This ppm assessment should be based on inspections performed on each inspection lot to verify that resistors meet dc resistance and tolerance requirements (i.e., ppm-2). For ER resistors, this inspection shall occur after the group A, subgroup 2, 100 percent screens have been completed. In the event of one or more failures, the lot shall be rejected.

4.6.4.1 Sampling plans. Minimum sample sizes for inspection lots shall be selected in accordance with table X. For non-ER resistors, the sampling system and plan used for the group A inspection (see 4.6.3.1) may be the basis for assessing ppm.

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4.6.4.2 Rejected lots. Any rejected inspection lot shall be segregated from new lots and those lots which have passed the ppm assessment. The manufacturer has two options on an rejected inspection lot.

Option 1:

a. The manufacturers can submit individual production lots that do not contain the samples that failed the inspection lot test to the ppm assessment. A sample from these production lots shall be selected in accordance with table X. Those production lots that have zero defects are available for shipment. Data from these production lots shall also be used for ppm calculations.

b. For those production lots that contain the samples that failed the inspection lot test, the production lot may be rescreened for the quality characteristics found defective in the sample and any defects removed. A second sample in accordance with table X shall be selected. If one or more defects are found, this lot is rejected and shall not be supplied to the specification.

Option 2:

a. A rejected inspection lot may be rescreened for the quality characteristics found defective in the inspection lot sample and any defects removed. A new second sample in accordance with table X shall be randomly selected. If one or more defects are found, this lot is rejected and shall not be supplied to this specification.

b. A rejected lot may be rescreened for the quality characteristics found defective in the sample and any defects removed. A new second sample shall be randomly selected. If one or more defects are found, this lot is rejected and shall not be supplied to the specification.

4.6.4.3 PPM calculations. PPM calculations shall be based on the accumulated results of the initial sample. Calculations and exclusions shall be in accordance with EIA-554-1. (NOTE: PPM calculations shall not be based on the second sample submission for a rejected lot as described in 4.6.4.2.).

4.6.5 Group B inspection (ER and space level only). Group B inspection shall consist of the inspections specified in table XI in the order shown, and shall be performed on samples from lots which have been subjected to and have passed the group A inspection. Group B inspection shall be performed on inspection lot basis for ER and production lot for space level.

4.6.5.1 Sampling plan.

4.6.5.2 Subgroup 1. Twenty sample units of parts shall be randomly selected from the inspection lot for ER and from the production lot for space level. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defectives, a new sample of 20 parts shall be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

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TABLE XI. Group B inspection (ER and space level).

Inspection	Requirement paragraph	Method paragraph	Number of samples
Subgroup 1 1/			
Resistance temperature characteristic	3.16	4.8.10	
Short time overload	3.12	4.8.6	20 samples
Subgroup 2 2/			
Solder mounting integrity 3/	3.19.1	4.8.13.1	10 samples
Bondable mounting integrity 4/	3.19.2	4.8.13.2	10 samples
Wire bonding integrity 5/	3.19.3	4.8.13.3	10 samples

1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity can be performed on a annual basis. If the design, material, construction, or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, this test, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every three years after the deletion as part of long term design verification. If the design, material, construction, or processing of the part is changed or, if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.

3/ Applicable to termination B.

4/ Applicable to termination's C, D, G, T, and U.

5/ Applicable to termination's W and P.

4.6.5.3 Subgroup 2. A sample of ten parts shall be randomly selected from the inspection lot for ER and from the production lot for space level. If one or more defects are found, the lot shall be reworked or screened and defectives removed. After reworking or screening and removal of defectives, a new sample of ten parts shall be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.5.3.1 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.7 Periodic group C inspection (ER only). Group C inspection shall consist of the tests specified in table XII, in the order shown. Group C inspection shall be made on sample units selected from lots which have passed group A and group B inspection. Group C samples shall be representative of production. The allowable number of defectives shall be as indicated in table XII.

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TABLE XII. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Monthly Subgroup 1				
Life (ER, ceramic substrate)	3.17.2	4.8.11.1	4.7.1.1	4.7.1.1
Subgroup 2 1/				
Thermal shock Low temperature operation	3.9 3.11	4.8.3 4.8.5	30	1
Quarterly Subgroup 1 1/				
Resistance to bonding exposure Moisture resistance	3.14 3.15	4.8.8 4.8.9	30	1
Subgroup 2 1/				
High temperature exposure	3.13	4.8.7	30	1
Subgroup 3				
Life (Fiberglass board)	3.17.2	4.8.11.2	15 (5 high, 5 critical, 5 low)	1

1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity, can be performed on a annual basis. If the design, material, construction, or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

4.7.1 Sampling plan. If more than 1,000 resistors of any style or style grouping are produced over the maintenance period, the group C tests shall be performed as specified. If the production rate is less than 1,000 resistors for any style or style grouping over the maintenance period, then the monthly or quarterly group C inspection may be postponed until at least 1,000 resistors of that style or grouping are produced (except for the monthly life test). In any case, the monthly tests shall be performed at least once every 3 months. The quarterly tests shall be performed at least every 6 months. This requirement is waived if the manufacturer has obtained a reduced inspection status through the qualifying activity.

4.7.1.1 Subgroup 1. Samples shall be accumulated from inspection lots and placed on extended life test in accordance with 4.8.11 once a month for the full 10,000 hour life test. A sufficient number of samples shall be selected by the manufacturer so that the maintenance of FR requirements comply with the specified maintenance period. Samples shall be representative of production of individual styles, characteristics, and termination's in the maintenance period. The manufacturer has the option, with qualifying activity approval, of selecting samples from RTC groups versus individual characteristics (group I: E or H; group II: K or M). As far as practical, the manufacturer shall select the resistance values so that all resistance decades produced during the maintenance period are represented.

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4.7.1.2 Subgroup 2. Subgroup 2 tests shall be performed on a sample size of 30 units of each style, characteristic, and termination with one defective allowed for each group of 30.

4.7.2 Quarterly.

4.7.2.1 Subgroup 1. Every 3 months, 30 units of each style, characteristic, and termination material (any resistance value) shall be subjected to quarterly tests specified in table XII, in the order shown.

4.7.2.2 Subgroup 2. Every 3 months, 30 units of each style, characteristic, and termination material (any resistance value) shall be subjected to quarterly tests specified in table XII, in the order shown.

4.7.2.3 Subgroup 3. Every 3 months, 15 units of each style, characteristic, and termination material, except W and P, shall be subjected to the subgroup 3 quarterly tests as specified in table XII. As an option to performing testing of each characteristic and termination finish, the manufacturer may construct groups based on the guidance in table XVII. (NOTE: Combination of styles is not permitted.)

4.7.3 Noncompliance. If a sample fails to pass group C inspection, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A and group B inspections may be reinstituted; however, final acceptance shall be withheld until group C reinspection has shown that the corrective action was successful

4.8 Methods of inspection.

4.8.1 Visual and mechanical inspection and chip mounting.

4.8.1.1 Visual and mechanical inspection. Unless otherwise specified, resistors shall be examined under 30X to 60X magnification. In case of conflict 30X will be the referee power, unless otherwise specified, to verify that the materials, design, construction, physical dimensions, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.23 and Appendix B).

4.8.1.2 Chip mounting procedures. When specified herein, the chip resistor shall be mounted on a test board as described in 4.8.1.3. For those test procedures where mounting requirements are unspecified, the chip resistors may be tested unmounted using pressure contacts.

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4.8.1.3 Specified mounting. When specified in the test procedure, the chip resistor shall be mounted on a suitable test board as specified on figure 3. The test board material shall be such that it shall not be the cause of, nor contribute to, any failure of a chip resistor in any of the tests for which it may be used. The test board shall be prepared with metallized surface land areas of proper spacing so that a test board surface area of at least four times the resistor chip surface area is provided for each resistor chip mounted. The metallized surface land areas shall be designed in a pattern to accommodate a number of resistor chips. The metallization material shall be compatible with the bonding technique to be employed and the material used on the chip termination. The method of chip mounting for the different termination materials shall be as follows:

a. All termination's (except P and W). All chip resistors (except P and W) termination materials shall be mounted on the test board by soldering the chip termination's directly to the test board metallized land areas in the following manner:

   (1) Solder and soldering flux shall be of such a quality as to enable the chip resistors to meet all the requirements of this specification and shall be applied to the termination's of each chip of code letters B, C, D, T, and U (see 1.2.4).

   (2) All chips shall be placed across the metallized land areas of the test board with contact between the chip termination's and board land areas only. The use of adhesive to keep chips in place during mounting operation is allowable except for preparation of solder mounting integrity test pieces.

   (3) For resistance to bonding exposure, the test board with all chips in position shall be placed on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with the temperature maintained at 260ºC ±10ºC for 5 seconds ±2 seconds after reflow.

   (4) For load life, the test board with all chips in position shall be placed on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with suitable time and temperature to assure proper reflow mounting.

b. Termination material "W" and "P" (wire bondable). Chip resistors with wire bondable termination's shall be mechanically attached to the test board and electrically connected by wire bonding flying lead interconnections between the chip termination and the test board metallized land areas. The interconnecting lead shall be .001 inch (0.03 mm) in diameter gold wire. The chip shall be mechanically or adhesively mounted by any procedure which will not be the cause of or contribute to any failures of the chip resistor in any test.

c. Alternative mounting for termination's D, C, T, and U. As an alternative to 4.8.1.3a(1), chip resistors may be mechanically attached to the test board with type 2 epoxy in accordance MMM-A-1931. The epoxy shall be cured at a minimum temperature of 150ºC. The minimum curing time shall be 30 minutes.

4.8.1.3.1 Fiberglass base board. Fiberglass base boards can be FR-4, G-10, G-30, or equivalent. When specified, a fiberglass test board with copper lamination shall be used for testing. The copper lamination shall be 70 micrometers copper (2 oz.) thickness (maximum).

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4.8.2 DC resistance (see 3.8). The dc resistance shall be measured in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same type, model, or if it can be shown that the performance of the equipment is equivalent or better.

b. Test voltage for bridges: Measurements of resistance shall be made by using the test voltages up to the maximum values specified in table XIII. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement was made.

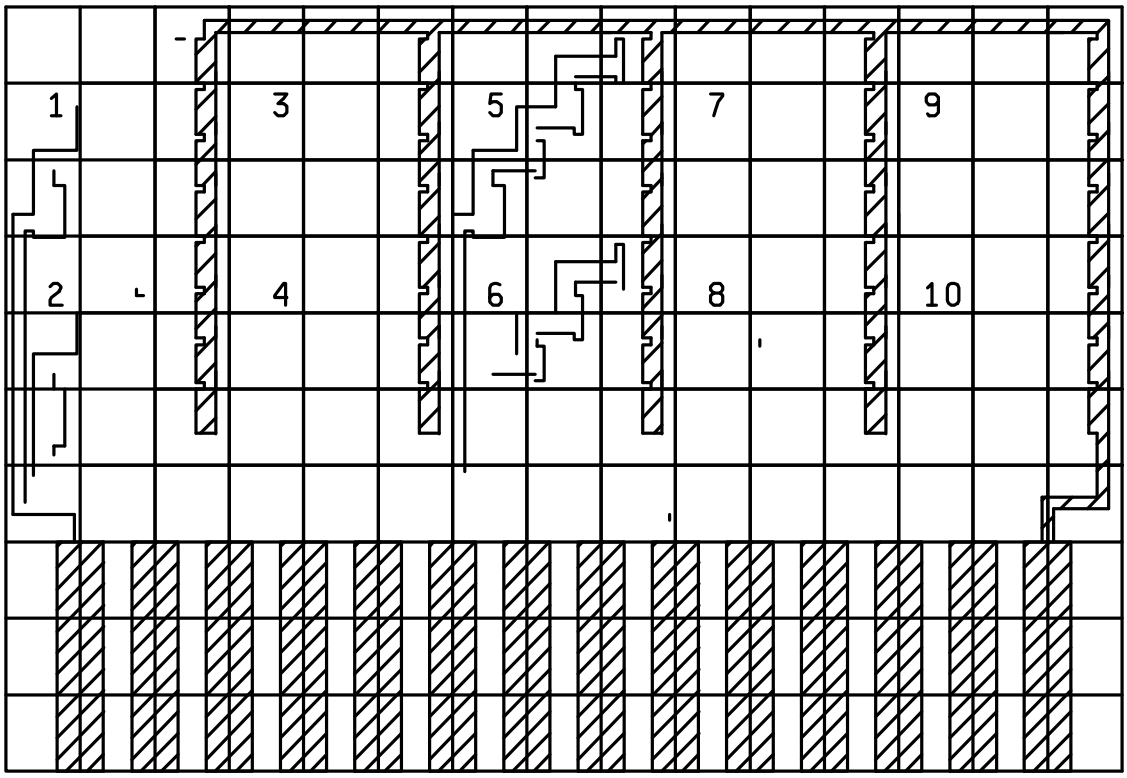
c. Measurement energy for electronic test equipment: The measurement energy applied to the unit under test shall not exceed 10 percent of the 25ºC rated wattage times 1 second.

d. Temperature: The dc resistance test specified in group I of table VIII shall be performed at 25ºC ±2ºC. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be within ±2ºC of the temperature at which the initial resistance measurement was made.

TABLE XIII. Standard dc resistance test voltages.

Nominal resistance	Maximum test voltage			
	10 to 24 milliwatts	25 to 99 milliwatts	100 to 225 milliwatts	226 to 1,000 milliwatts
ohms	Volts	Volts	Volts	Volts
5.60 to 9.88	0.2	0.3	0.7	1.0
10 to 98.8	0.3	0.3	1.0	1.0
100 to 988	1.0	1.0	1.0	3.0
1,000 to 9,880	3.0	3.0	3.0	10.0
10,000 to 98,880	10.0	10.0	10.0	30.0
100,000 or higher	30.0	30.0	30.0	100.0

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NOTES:
1. Suggested ceramic test board: 1.000 inch (25.4 mm) x 1.500 inch (38.10 mm) x 0.025 inch (0.64 mm) thick.
2. Suggested fiberglass test board: 1.000 inch (25.4 mm) x 1.500 inch (38.10 mm) x .0625 inch thick (1.5875 mm).
3. The contacts fit a printed circuit board connector with .100 inch (2.54 mm) in contact center.
4. Short lead wire may be soldered on and it will then fit a microcircuit connector, or test lead wires may soldered on for direct measurement.
5. Small grid square are .10 inch (0.25 mm).
6. A larger test board, to suitable scale, may be used to accommodate additional test samples or additional style/size configurations.

FIGURE 3. Suggested test board.

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4.8.3 Thermal shock (see 3.9). Resistors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions apply:

a. Mounting: Resistors shall not be mounted. Resistors may be placed in metal baskets, vials, or other apparatus as long as resistors are subjected to the specified temperature extremes. (For group C inspection only, as an option to the manufacturers, resistors may be mounted or unmounted.)

b. Measurement before cycling:

(1) Qualification and group A tests: DC resistance shall be measured as specified in 3.8, except at 25ºC ±5ºC. The dc resistance shall be within the specified resistance tolerance of the nominal resistance value.

(2) Group C test: DC resistance shall be measured as specified in 3.8.

c. Test condition F (except temperatures shall be 150ºC +10ºC, -0ºC and -65ºC +0ºC, -10ºC).

d. Measurement after cycling:

(1) Qualification and group A tests: After stabilization at room temperature, the dc resistance shall again be measured as specified in 3.8, except at 25ºC ±5ºC.

(2) Group C test (not applicable for space level): After stabilization at room temperature, the dc resistance shall again be measured as specified in 3.8. Following the test, the resistors shall be examined for evidence of mechanical damage.

4.8.4 Power conditioning (space level only) (see 3.10). Chip resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Test temperature and tolerance: 70ºC +5ºC, -0ºC.

b. Initial measurements: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured, at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the chip resistors to the test temperature. This initial measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured at room temperature. This measurement shall be used as the reference temperature for all subsequent measurements under the same condition.

c. Operating condition: A dc voltage or filtered or nonfiltered full-wave rectified ac voltage shall be applied to the chip resistor intermittently, one and one-half hours on and one-half hour off, for 100 hours $\pm$4 hours and at the test temperature. During the “on” cycle, the voltage shall be regulated and controlled to maintain $\pm$5 percent of the required test voltage. Unless otherwise specified (see 3.1), power applied shall be one and one-half times rated power not to exceed maximum voltage for the style.

d. Measurements after test: Following a minimum one-half hour stabilization period, dc resistance shall be measured as specified in 3.8.

e. Inspection after test: chip resistors shall be inspected for evidence of mechanical damage.

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4.8.5 Low temperature operation (see 3.11). A dc resistance measurement shall be made as specified in 3.8, mounted on a ceramic test board as specified in 4.8.1.3. The resistors shall be placed in a cold chamber maintained at -65ºC +0ºC, -5ºC. After one hour of stabilization at this temperature, full rated continuous working voltage as specified in 3.7 shall be applied for 45 minutes +5 minutes, -0 minutes. Fifteen minutes +5 minutes, -0 minutes after the removal of the voltage, the resistors shall be removed from the chamber and maintained at a temperature of 25ºC ±5ºC until stabilized; the dc resistance shall be measured as specified in 3.8. Resistors shall be examined for evidence of mechanical damage.

4.8.6 Short time overload (see 3.12). The dc resistance shall be measured as specified in 3.8 and mounted on a ceramic test board as specified in 4.8.1.3. A dc test potential, 2.5 times the rated continuous working voltage but not exceeding twice the maximum voltage (see 3.1), shall be applied for 5 seconds to the resistor under the following conditions:

a. Resistors on test board shall be mounted so that the larger space of the resistor is on a horizontal plane.

b. In still air, with no circulation other than that created by the heat of the resistors being operated.

The resistors shall be stabilized at 25°C ±3°C after which the dc resistance shall be measured as specified. Resistors shall be examined for evidence of arcing, burning, and charring.

4.8.7 High temperature exposure (see 3.13). Resistors shall be mounted on a ceramic test board as specified in 4.8.1.3 and the dc resistance shall be measured as specified in 4.8.2. The resistors shall then be subjected to a continuous temperature of 150ºC ±5ºC for a period of 100 hours ±4 hours, in a test chamber with forced air circulation. The resistors shall be stabilized at 25ºC ±3ºC after which the dc resistance shall be measured. The change in resistance shall not exceed the value specified in 3.13.

4.8.8 Resistance to bonding exposure (see 3.14). The dc resistance of the chip resistor shall be measured as specified in 3.8. The chip resistor shall be mounted on a ceramic test board in accordance with 4.8.1.3. The test board, with resistors mounted, shall be stabilized at room temperature after which the dc resistance shall again be measured as specified in 3.8. The change in resistance shall not exceed the value specified in 3.14.

4.8.9 Moisture resistance (see 3.15). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Chip resistor sample units shall be mounted on a ceramic test board as specified in 4.8.1.3.

b. Initial measurement: Immediately following the initial conditioning period, dc resistance of each chip on the test board shall be measured as specified in 3.8.

c. Polarization and loading voltage: No voltage shall be applied.

d. Subcycle: Step 7b shall not be applicable. Step 7a shall be performed during any five of the first nine cycles only.

e. Measurements at high humidity: None.

f. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be removed from the chamber and air dried for 30 minutes ±15 minutes. The dc resistance shall then be measured in accordance with 3.8.

g. Examination after test: Resistors shall be examined for evidence of electrical and mechanical damage.

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4.8.10 Resistance temperature characteristic (see 3.16). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The following details and exceptions shall apply:

a. Reference temperature: Room ambient temperature.

b. Test temperature: In accordance with table XIV.

c. Accuracy of temperature measurement: Resistors shall be maintained for 30 minutes to 45 minutes within 3ºC at each of the test temperatures listed in table XIV. This tolerance shall be maintained on the established test temperatures.

d. Mounting: Resistors shall be mounted on a ceramic test board as specified in 4.8.1.3.

As an option, the manufacturer may develop a control procedure for minimum stabilization times required for RTC testing. This procedure shall detail specific time elements required to reach thermal and electrical stability for specific part designs. Upon approval by the qualifying activity, the manufacturer may use this procedure for minimum stabilization times for RTC testing.

TABLE XIV. Resistance temperature characteristic.

Sequence	Temperature
	Group B acceptance inspection 1/
	ºC
1	2/ 25 ±3
2	-55 ±3
3	2/ 25 ±3
4	125 ±3

1/ At the option of the manufacturer, the reverse sequence may be as follows:
1 - 1/ 25 ±3
2 - 125 ±3
3 - 1/ 25 ±3
4 - -55 ±3

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

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4.8.11 Life (see 3.17).

4.8.11.1 Life test (ceramic boards) (failure rate level system). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: Chip resistor sample units shall be mounted on a test board as specified in 4.8.1.3.

b. Test temperature and tolerance: 70ºC ±5ºC.

c. Initial resistance measurement of mounted resistors: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured after mounting at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the resistors to the test temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured after mounting at room temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

d. Operating conditions: Rated dc continuous working voltage, or filtered or nonfiltered full wave rectified ac voltage, shall be applied intermittently, 90 minutes on and 30 minutes off, for the applicable number of hours (see 4.8.11f) and at the applicable test temperature. "On time" shall be three quarters of the total elapsed time. During the "on" cycle, the voltage shall be regulated and controlled to maintain ±5 percent of the rated continuous working voltage.

e. Test condition: Two thousand hours elapsed time for qualification inspection with all samples continued to 10,000 hours. Ten thousand hours for FRL inspection group C.

f. Measurements during test:

(1) Qualification inspection: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours; 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -24 hours have elapsed.

(2) Extended life testing: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours, 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -24 hours, and every 2,000 hours +96 hours, -24 hours thereafter until the required 10,000 +120, -0 hours have elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted so that measurements need not be made during other than normal workdays.

(3) Measurements outside of the chamber: When measurements are made outside the chamber, resistors shall be outside of the chamber, for a minimum of 45 minutes and stabilized before measurement.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

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4.8.11.2 Life test (fiberglass board) (see 4.8.1.3.1). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: Chip resistors sample units shall be mounted on a test board as specified in 4.8.1.3.

b. Test temperature and tolerance: 70ºC ±5ºC.

c. Initial resistance measurement of mounted resistors: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured after mounting at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the resistors to the test temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured after mounting at room temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

d. Operating conditions: Rated dc continuous working voltage, or filtered or nonfiltered full wave rectified ac voltage, shall be applied intermittently, 90 minutes on and 30 minutes off, for the applicable number of hours (see 4.8.11f(1)) and at the applicable test temperature. "On time" shall be three quarters of the total elapsed time. During the "on" cycle, the voltage shall be regulated and controlled to maintain ±5 percent of the rated continuous working voltage.

e. Test condition: Two thousand hours elapsed time for qualification inspection with all samples continued to 10,000 hours. Ten thousand hours for FRL inspection group C.

f. Measurements during test:

(1) Qualification inspection: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours; 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -24 hours have elapsed.

(2) Extended life testing: DC resistance shall be measured at the end of the 30 minutes off periods after 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours, 1,000 hours +72 hours, -24 hours; and 2,000 hours +96 hours, -24 hours, and every 2,000 hours +96 hours, -24 hours thereafter until the required 10,000 +120, -0 hours have elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted so that measurements need not be made during other than normal workdays.

(3) Measurements outside of the chamber: When measurements are made outside the chamber, resistors shall be outside of the chamber, for a minimum of 45 minutes and stabilized before measurement.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

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4.8.12 Solderability (applicable to termination B) (see 3.18). Resistors shall be tested in accordance with method 208 of MIL-STD-202. Both end termination's shall be immersed completely one at a time or both at the same time by dipping the entire chip in the solder pot. The following details and exceptions shall apply:

a. Apparatus:

(1) A solder pot capable of keeping the solder temperature to ±5ºC of setpoint.

(2) A device to keep the specimen submerged in solder for the required time.

(3) Optical equipment capable of 30X magnification.

(4) A laboratory beaker or appropriate laboratory apparatus that is capable of:

(a) Supporting a nonmetallic device.

(b) Maintaining the test specimens at a distance of 1.5 inches (38.1 mm) to 2.5 inches (63.5 mm) from the surface of the boiling deionized water (DI) or distilled water while, at the same time, exposing them to the full flow of steam generated.

b. The requirement for standard solderable wire application shall not apply.

c. The immersion and emersion rates shall not apply.

d. The specimen shall be totally submerged for at least 5 seconds.

4.8.13 Mounting integrity.

4.8.13.1 Solder mounting integrity (termination B). The resistor chips shall be prepared as specified in 4.8.1.3a. A force shall be applied to all solderable units. For styles RM0502, RM0505, RM1005, RM1505, RM0705, and RM1206, the force applied shall be 2 kilograms. For styles RM2208, RM2010, RM2512, and RM1010 the force applied shall be 3 kilograms. The force shall be applied to the edge of the chip as shown on figure 4 for a minimum of 30 seconds. The resistor shall be examined for evidence of mechanical damage. The pusher width shall be a minimum of 30 percent of the length and the maximum width shall be 70 percent of the chip length.

4.8.13.2 Bondable mounting integrity (termination's C, D, G, T, and U). The resistor chips shall be prepared as specified in 4.8.1.3c. A force shall be applied to all bondable units. For styles RM0502, RM0505, RM1005, RM1505, RM0705, and RM1206, the force applied shall be 1 kilogram (for termination D and T the applied force shall be 0.75 kilograms). For styles RM2208, RM2010, RM2512, and RM1010, the force applied shall be 1.5 kilograms. The force shall be applied to the edge of the chip as shown on figure 4 for a minimum of 30 seconds. The resistor shall be examined for evidence of mechanical damage. The pusher width shall be a minimum of 30 percent of the resistor length and the maximum width shall be 70 percent of the chip length.

4.8.13.3 Wirebonding integrity. The chip resistor shall be prepared as specified in 4.8.1.3b. A pull of 4 grams shall be applied by inserting a hook under the lead wire at approximately the center of the wire as shown in figure 5. The force shall be applied at a 90 degree angle to the surface of the chip, one lead at a time for a minimum of 30 seconds each.

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4.8.14 Resistance to solvents (see 3.20). Resistors shall be tested in accordance with method 215 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Unmounted.

b. The marked portion of the resistor shall be brushed.

c. The number of sample units shall be as specified in table IX.

d. Resistors shall be inspected for mechanical damage and legibility of minimum marking.

4.8.15 Laser marking legibility test (see 3.21). Resistors shall be coated with .005 inch (0.13 mm) minimum of silicon resin insulated compound, type SR of MIL-I-46058. After curing, coated resistors shall be examined for legibility under normal production room lighting by an inspector at 10X magnification.

4.8.16 Outgassing (see 3.24). Resistors shall be tested in accordance with ASTM E595. The following details and exceptions shall apply. Samples of any materials used shall be cut into pieces having .38 inch (9.6 mm) maximum dimension. A sample of 100 milligrams to 300 milligrams of material shall be weighed and then placed in a vacuum chamber under the following conditions:

Pressure ---------- 10^{-6} torr or less.
Temperature of specimen ------ Maximum operating temperature of device.
Exposure time ---------- 24 hours.

During this exposure, a collector plate 9.7 inches (246.4 mm) in diameter, shall be held at 25ºC $\pm$1ºC near the sample. Immediately upon removal of the specimen and collector plate from the chamber, they shall be weighed. The TML is sample mass before exposure minus the sample mass after exposure divided by the initial sample mass (expressed in percent change). The VCM is the increase in the weight of the collector plate divided by the mass of sample (expressed in percent).

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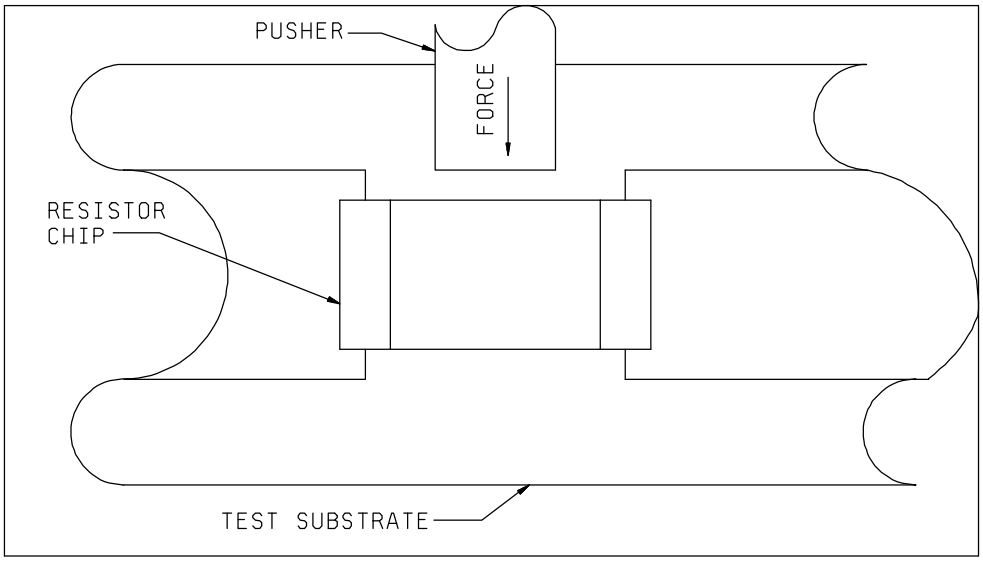


FIGURE 4. Mounting integrity test fixture.

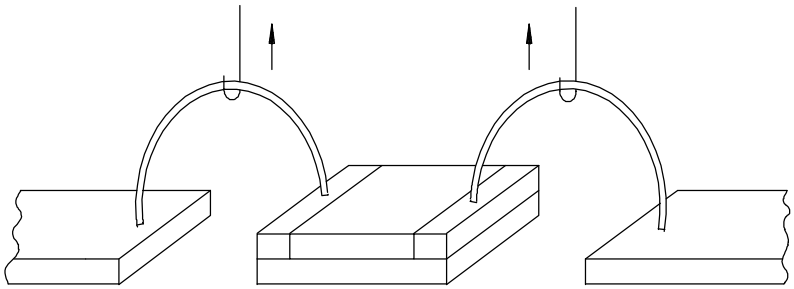


FIGURE 5. Weldable style units.

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Chip resistors are intended to be used in thin or thick film hybrid circuits where micro-circuitry is indicated and in surface mount applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, the applicable associated specification, and the complete PIN (see 1.2.1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

d. Allowable substitution (see 3.25.4 and 3.25.5).

e. If marking is required (see 3.25).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the QPL whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the QPL is the US Army Communications - Electronics Command and Fort Monmouth, ATTN: AMSEL-LC-LEO-E EP, Fort Monmouth, NJ 07703-5023; however, information pertaining to qualification of products may be obtained from the Defense Supply Center Columbus (DSCC-VQP), 3990 E. Broad Street, Columbus, OH 43216-5000.

6.4 Application notes.

6.4.1 Mounting of resistors. Resistors may be mounted individually on a test board, fiberglass base, and connected to conductor areas by means of solder preforms, conductive cement, or wire bonding. Users are cautioned that the substrate and chip component must have compatible temperature coefficients of expansion to reduce the strain level imposed on the solder connections. They may also be directly connected to other components on the same test board by means of wire bonding using the test board as a base or carrier for the resistor.

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6.4.2 Stacking of resistors. Stacking is generally discouraged since experience indicates that failure may occur due to electrolytic action in the bonding adhesive. This action occurs from the increase in temperature caused by stacking in conjunction with the operating voltage across the resistor. The bonding adhesive at the resistor chip interface chemically reacts with the active resistor material changing the noise and resistance values and eventually the resistor will open. If stacking resistors, care should be taken to compensate for the lower heat dissipation capabilities by derating the wattage rating.

6.4.3 Close tolerance resistors. Close tolerance resistors (i.e., ±0.1 percent) should be mounted on a test board by a method which produces the least heating effect over a short time to avoid permanent changes in resistance.

6.4.4 Resistor film orientation. It is recommended that the film side of resistors be mounted up when they are placed onto printed circuit boards. This is due to the effect of thermal stressing on epoxies used to hold them in place when mounted on the bottom of printed circuit boards. Resistor films can be torn from the substrate during thermal cycling, substantially altering the value of the resistor.

6.4.5 Tin plated finishes. Tin plating is prohibited (see 3.5.3) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545.

6.5 Noise level test. This specification does not include a noise test. If the user requires a noise test, it should be performed in accordance with method 308 of MIL-STD-202.

6.6 Electrostatic charge effects. Under relatively low humidity conditions, some types of film resistors, particularly those with small dimensions and high sheet resistivity materials, are prone to sudden significant changes in resistance (usually reductions in value) and to changes in temperature coefficient of resistance as a result of discharge of static charges built up on associated objects during handling, packaging, or shipment. Substitution of more suitable implements and materials can help minimize this problem. For example, use of cotton gloves, static eliminator devices, air humidifiers, and operator and work bench grounding systems can reduce static buildup during handling. Means of alleviating static problems during shipment include elimination of loose packaging of resistors and use of metal foil (conductive) and static dissipative packaging materials.

6.7 Subject term (key word) listing.

Failure rate
Surface mount
Termination

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX A

PROCEDURES FOR QUALIFICATION INSPECTION

A.1 SCOPE

A.1.1 Scope. This appendix covers the procedure for qualification inspection of resistors covered by this specification. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance only. This appendix also outlines the procedure for extending qualification to the various product levels as well as the resistance range, characteristics, tolerances, and FRLs. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS. This section is not applicable to this document.

A.3 SUBMISSION

A.3.1 Product levels. Qualification of the C (non-ER) level, is predicated upon meeting the ER qualification requirements for FRL P (see A.4.1). Qualification of the T (space) level, is predicated upon meeting the ER qualifications requirements for FRL R or S (see A.4.1). To obtain qualification to the T (space) level, the manufacturers shall also meet the additional requirements for outgassing and the 100 cycle thermal shock test (see A.3.2) as well as been approved by the qualifying activity on the capability to conduct tests and examinations for the space level product (e.g., resistance noise, destructive physical analysis, production lot formation). The procedure for submitting samples to become qualified to the initial FRL M is specified in A.3.2.

A.3.2 Sample. A sample consisting of 222 sample units, 74 of the lowest, 74 of the highest, and 74 of the critical or nearest to critical value (see table XV), and in each style and characteristic, termination material, and resistance tolerance for which qualification is sought, shall be submitted and subjected to the inspection of table VIII. If no critical value is specified, the sample will be equally divided between the highest and lowest resistance values. Ten sample units of any resistance value or characteristic shall be submitted and subjected to the test of group V. Twenty sample units of any resistance value shall be submitted to tests of group VI, when applicable. Thirty sample units shall be submitted to the test in group VII. If qualification to a closer resistance tolerance than submitted above is desired, 30 additional sample units of the closer resistance tolerance shall be subjected to the following tests (see table XVI). After qualification has been granted, no changes shall be made in materials, design, or construction without prior notification to the qualifying activity.

TABLE XV. Sample size for qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample units for inspection	Number of defectives allowed
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, 3.23 and 3.25	4.8.1	30 (10 highest, 10 critical, 10 lowest)	0
DC resistance	3.8	4.8.2		
Thermal shock	3.9	4.8.3		

A.3.1.1 Qualification to extended resistance ranges. For qualification to the extended resistance ranges of higher resistance tolerances, 74 additional samples of high or low resistance shall be subjected to the tests of group I, group II, group III, and group IV of table VIII.

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APPENDIX A

TABLE XVI. Critical resistance values for qualification ranges. 1/

Style	Slash sheet	Critical resistance value 2/ 3/							
		Characteristic							
		E		H		K		M	
		Ceramic board	Fiber board	Ceramic board	Fiber board	Ceramic board	Fiber board	Ceramic board	Fiber board
RM0502	/1	158.0	158.0	158.0	158.0	78.7	78.7	78.7	78.7
RM0505	/2	63.4	63.4	63.4	63.4	31.6	31.6	31.6	31.6
RM0705	/6	49.9	49.9	49.9	49.9	24.9	24.9	24.9	24.9
RM1005	/3	31.6	31.6	31.6	31.6	15.8	15.8	15.8	15.8
RM1010	/10	22.1	22.1	22.1	22.1	11.0	14.0	11.0	14.0
RM1505	/4	15.8	15.8	15.8	15.8	10.5	10.5	10.6	10.6
RM2208	/5	7.87	7.87	7.87	7.87	6.98	6.98	6.98	6.98
RM1206	/7	78.7	78.7	78.7	78.7	39.2	39.2	39.2	39.2
RM2010	/8	56.2	56.2	56.2	56.2	----	----	----	----
RM2512	/9	78.7	78.7	78.7	78.7	39.2	52.3	39.2	52.3

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when the maximum continuous working voltage is applied.
3/ All resistance in kilohms.

A.4 EXTENT OF QUALIFICATION

A.4.1 Extent of qualification. The resistance range included in the qualification of any one style and tolerance shall be between any two adjacent resistance values which pass the required qualification inspection. The extent of qualification between styles, characteristics, termination's, FRLs, and resistance tolerances shall be as specified in table XVII.

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TABLE XVII. Extension of qualification.

Style	Will qualify style 1 /	
RM0502	RM0502	
RM0505	RM0505	
RM0705	RM0705, RM0505	
RM1005	RM1005, RM0705, RM0505	
RM1010	RM1010, RM1005, RM0705, RM0505	
RM1505	RM1505, RM1005, RM0705, RM0505	
RM2208	RM2208	
RM1206	RM1206, RM1505, RM1010, RM1005, RM0705, RM0505	
RM2010	RM2208, RM2010	
RM2512	RM2010, RM2208, RM2512	
Characteristic	Will qualify characteristic	May qualify characteristic with minor additional testing 2/
E	E, H	K, M
H	H	K, M
K	K, M	
M	M	
Termination	Will qualify termination	May qualify termination with minor additional testing 2/
G	G, W, P	B
W	W, P	
P	P, W	
B	B	G, W, P 3/
U	U, T	C, D
T	T	D
C	C, D	U, T
D	D	T
Product level	Will qualify product level	
T	R, P, M, C	
S	P, M, C	
R	M, C	
P	C	
M		
C		
Resistance tolerance	Will qualify resistance tolerance	
0.1	0.1, 1.0, 2.0, 5.0, 10.0	
1.0	1.0, 2.0, 5.0, 10.0	
2.0	2.0, 5.0, 10.0	
5.0	5.0, 10.0	
10.0	10.0	

1/ The same design and technology must be utilized for the style qualified and the styles that the qualification is extended to.
2/ To be determined by qualifying activity.
3/ Must contain barrier layer to qualify.

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APPENDIX B

VISUAL INSPECTION

B.1 SCOPE

B.1.1 Scope. This appendix details the procedure for visual inspection of conductors, metallization, scratches, voids, corrosion, substrate defects, etc., of chip resistors covered by this specification. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance only.

B.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

B.3 INSPECTION

B.3.1 Conductor metallization defects (thick or thin film). Any resistor which exhibits the following defects in the active circuit metallization shall not be acceptable. The active circuit metallization is all metal or any other material used for interconnection except metallized scribe lines, unused bonding pads, and identification marking.

B.3.1.1 Conductor metallization scratches (thick or thin film). A scratch is defined as any tearing defect that disturbs the original surface of the metallization.

a. Any scratch in metallization through which the underlying resistor material also appears to be scratched.

b. Any scratch in the interconnecting metallization which exposes resistive material, substrate, or oxide anywhere along its length and reduces the width of the connecting strip to less than 50 percent of its original width.

B.3.1.2 Conductor metallization voids (thick or thin film). A void is defined as any region in the interconnecting metallization where the underlying resistive material or substrate is visible which is not caused by a scratch. Any void in the interconnecting metallization which leaves less than 50 percent of the original width undisturbed (see figure 7).

B.3.1.3 Conductor metallization nonadherence (thick or thin film). Conductor metallization nonadherence is defined as any evidence of metallization lifting, peeling, or blistering (see figure 7).

B.3.1.4 Conductor metallization bridging (thin film) (see figure 7). Conductor metallization bridging defects are defined as follows:

a. Bridged metallization defect that reduces the distance between any two metallization areas to less than 50 percent or 0.3 mils, whichever is greater.

b. Bridging between metallization and resistor pattern not intended by design that reduces the distance between the two to less than 0.1 mils.

B.3.1.5 Conductor metallization missing (thick or thin film). Conductor metallization defects are defined as more than 5 percent of the metallization area missing from the bonding pad (see figure 7).

B.3.1.6 Conductor metallization corrosion (thick and thin film). Metallization corrosion is defined as any evidence of localized heavy stains, metallization corrosion, discoloration or mottled metallization (see figure 7).

B.3.1.7 Excessive metallization (thick and thin film). A buildup or protrusions of termination material shall not exceed 0.003 inches (0.08 mm) above the average thickness of the termination's surface (see Figure 7).

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B.3.2 Resistor defects (thick and thin film). Any resistor which exhibits the following defects in the active area shall not be acceptable. The active area of a resistor is that part of a resistor pattern which remains in series connection between resistor terminals and is not shorted by metallization. For resistor patterns using laser trimmed "rungs", any anomaly in a non-active area at least two rungs beyond the last properly trimmed rung are acceptable.

B.3.2.1 Resistor scratches (thick or thin film) (see figure 8). Resistor scratches are defined as any scratch within the active resistor area one mil or longer. A scratch must be indicated by either disturbance of the surface resistor or extraneous, shiny material remaining from the instrument causing the scratch.

B.3.2.2 Resistor voids (thick and/or thin film) (see figure 8 and figure 10).

a. For block patterns, a void or neckdown in the active resistor area which reduces the width of the area by more than 10 percent of the original area (see figure 8).

b. For thin film serpentine and link patterns a void or neckdown in the active resistor area which reduces the width of the area by more than 50 percent of the original area (see figure 10).

B.3.2.3 Resistor nonadherence (thick and thin film). Resistor nonadherence is defined as any evidence of resistor film lifting, peeling, or blistering (see figure 8 and figure 10).

B.3.2.4 Resistor cracks (thick and thin film). Any chip out or crack in the resistor area.

B.3.2.5 Resistor material corrosion (thick and /or thin film). Resistor material corrosion is defined as follows:

a. Any evidence of localized heavy stains or corrosion of resistor material in the active resistor area (see figure 8 and figure 10).

b. For thin film, discoloration of the resistor film due to thermal stabilization is not a cause for rejection.

B.3.2.6 Resistor bridging defects (thin film). Resistor bridging defects are defined as follows (see figure 10):

a. Resistor bridging caused by photolithographic defect that forms a conductive bridge across two or more adjacent active resistance areas where width of bridge path is less than one half the smallest line width in the pattern.

b. Partial bridging defect that reduces the distance between adjacent active resistance areas to less than 50 percent of the design separation.

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B.3.2.7 Resistor trim faults (thick or thin film). After trimming any resistor which exhibits the following defects in the active area shall not be acceptable (see figure 6 and figure 9).

Applies to thin film:

a. The narrowest remaining width after trimming must be equal to or greater than 50 percent of the width of the narrowest line within the same resistor pattern (see figure 9).

b. For a resistor pattern with links designed for incremental trim where edge of cut touches the active resistor area (see figure 9).

c. Discoloration of resistors in and around laser kerf is not cause for rejection.

Applies to thick and thin film:

d. Uncut resistor material remaining after trimming due to "skipping" (see figure 6).

e. Voids, cracking, or similar damage caused to the underlying material by trimming where such damage touches active interconnects, resistor area, or both.

f. Any unintentional "scorching" of resistor areas by direct laser beam or spurious reflections caused by optics of the system.

g. For a cut in the resistor pattern which reduces a link or ladder width by more than 50 percent of original width. Links cut 100 percent are acceptable (see figure 9).

B.3.3 Substrate defects (thick and/or thin film). Any resistor which exhibits a substrate defect is cause for rejection and defined as follows (see figure 6 and figure 9):

a. Any crack that exceeds 3.0 mil in length or comes closer than 1.0 mil to an active resistor or bonding pad area on the substrate (see figure 6 and figure 9).

b. Any crack in a substrate that exceeds 1.0 mil in length and points towards the active resistor area (see figure 6 and figure 9).

c. A substrate, when scribed and broken, is left with an attached portion of an adjacent substrate which contains metallization or resistor material.

d. Any crack not originating at a substrate edge is cause for rejection (see figure 6 and figure 9).

B.3.4 Foreign material (thick and/or thin film). Material is considered attached when it cannot be removed with a soft bristled brush or a nominal gas blow (of 20 psig) of dry nitrogen or air. Attached foreign material which reduces the separation between metallization areas to less than 0.1 mil or is greater than 5 percent of the surface area on which it occurs is cause for rejection.

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B.3.5 Heavy metal marks (thick and thin film). Metal marks are defined as termination material on the surface of the substrate or passivation. Heavy metal marks are defined as dots of solder that are continuous and do not allow the substrate/passivation to be seen between the individual dots of metal when viewed at 30X. Light metal marks are defined as dots of metal when viewed at 30X. Light metal marking can be accepted.

a. Reject for any heavy metal marks greater than 10 mils in any direction.

b. Reject for 5 or more heavy metal marks larger than 3 mils but less than 10 mils in any direction on a surface of a resistor chip.

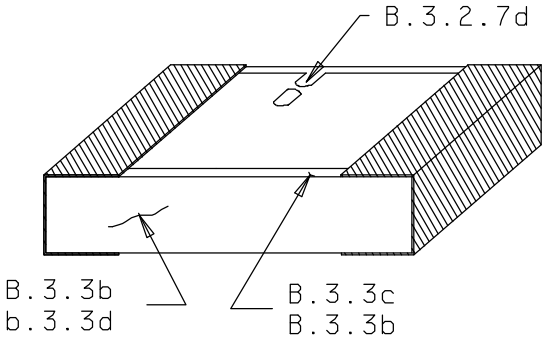


FIGURE 6. Cracks, chip outs, or voids.

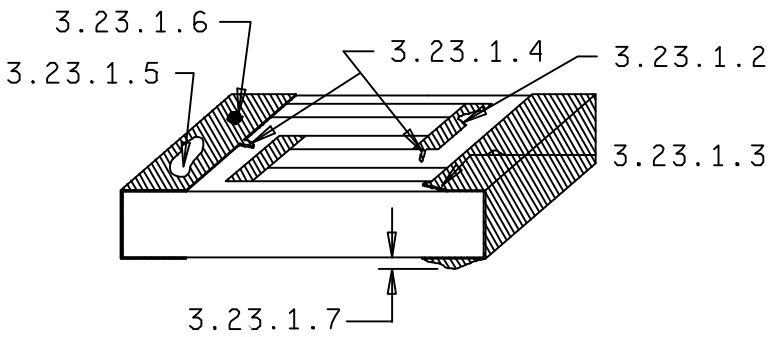


FIGURE 7. Conductor metallization defects.

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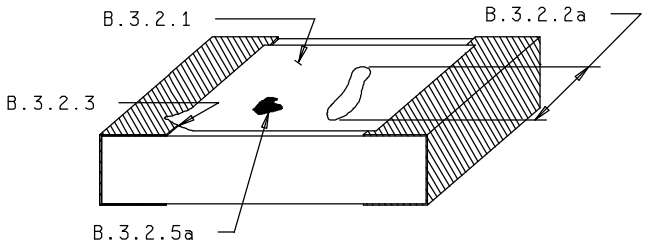


FIGURE 8. Resistor voids.

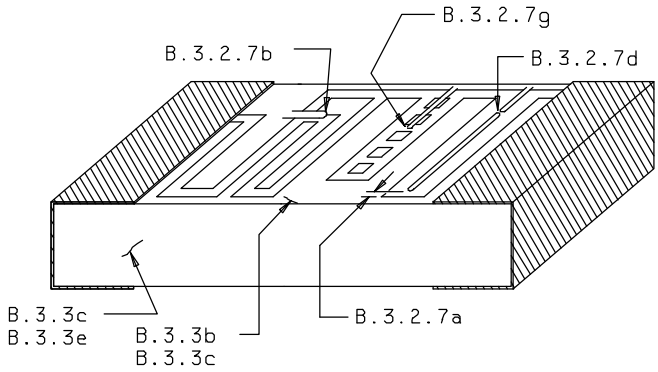


FIGURE 9. Resistor trim faults and substrate defects.

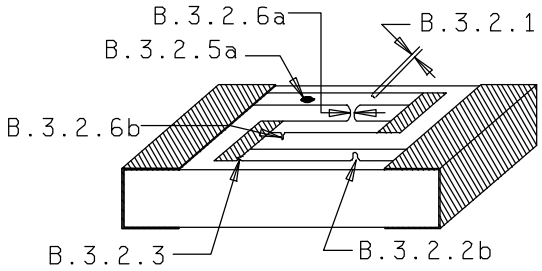


FIGURE 10. Resistors defects.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, CR4
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5905-1459)

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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3. DOCUMENT TITLE RESISTORS, FIXED, FILM, CHIP, NON-ESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, SPACE LEVEL, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

NAME US Army Communications and Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9104 (2) AUTOVON 992-9104
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